

# Ahmed Mamdouh

## PROFILE

Software engineer with professional experience in enterprise software and computer vision/AI. Built a strong C/C++ foundation through Computer Science & Engineering studies and a High-Performance Computing minor covering modern C++17/20, RAII, resource management, and performance-oriented programming. Seeking C++ and systems programming roles.

## EXPERIENCE

### Software Developer, SAP Pioneer; Walldorf/Remote – 2024-2026

- Built and maintained features for the Financial Products Subledger (FPSL).
- Designed and optimized high-performance ABAP SQL queries and data-oriented processing for financial applications.
- Contributed to architecture discussions, pair programming, code reviews, and modernization of internal developer and quality tooling.

### CV/ML Student Developer, Primetals Technologies; Erlangen – 2023-2024

- Built AI-powered assistants and computer-vision pipelines for real-time industrial workflows in steel mills.
- Trained and fine-tuned computer-vision models and hardened them into production-ready pipelines.
- Cut inference latency on an existing pipeline from -1,000ms to 2ms – a 500x speedup that enabled real-time use.
- Extended reusable internal ML tooling for future industrial AI projects.

### Junior Software Developer, Sequel Solutions; Cairo, Egypt – 2020-2021

- Built web and mobile features for a vacation-booking platform with JavaScript, React Native, Firebase, Docker.
- Modernized legacy code, cutting page-load time by 30% and reaching a perfect Lighthouse score.
- Added backend caching and contributed mobile-specific functionality.

### Perception Lead, Autonomous Bike Team @ GUC; Cairo, Egypt – 2020-2021

- Led research and implementation of monocular depth-perception pipelines for an autonomous bicycle
- Evaluated computer-vision approaches for estimating scene depth from a single camera under real-world constraints.

## PROJECTS

### Eden – Modern C++20 Graphics Library / Engine Project

- Building a modular graphics library/engine in C++20 with Vulkan to deepen hands-on experience with low-level graphics and systems programming.
- Applying modern C++ practices including RAII, explicit resource ownership, reusable abstractions, CMake, and performance-conscious design.

### Askier – C++ Real-Time Image Processing Application

- Contributed to an existing image-processing application with GPU-acceleration and parallel processing.
- Implemented custom OpenCL kernels for image-processing and integrated additional processing through OpenCV.

## EDUCATION

Friedrich-Alexander-Universität, Erlangen – M.Sc. AI – Minor in High-Performance Computing, 2025

German University in Cairo, Cairo, Egypt – B.Sc Computer Science & Engineering, 2021

## SKILLS

- **C++** – C++17/20 · C · CMake · Qt · OpenGL · Vulkan · OpenCV · OpenCL · GoogleTest
- **AI / Computer Vision** – Python · PyTorch · ONNX · OpenCV · ML pipelines · RAG / Agentic workflows
- **Software Engineering** – OOP · Software architecture & design · Data structures and algorithms · Distributed Systems · REST APIs · SQL · Git · CI/CD · Docker
- **Additional** – JavaScript/TypeScript · React · React Native · Java · ABAP